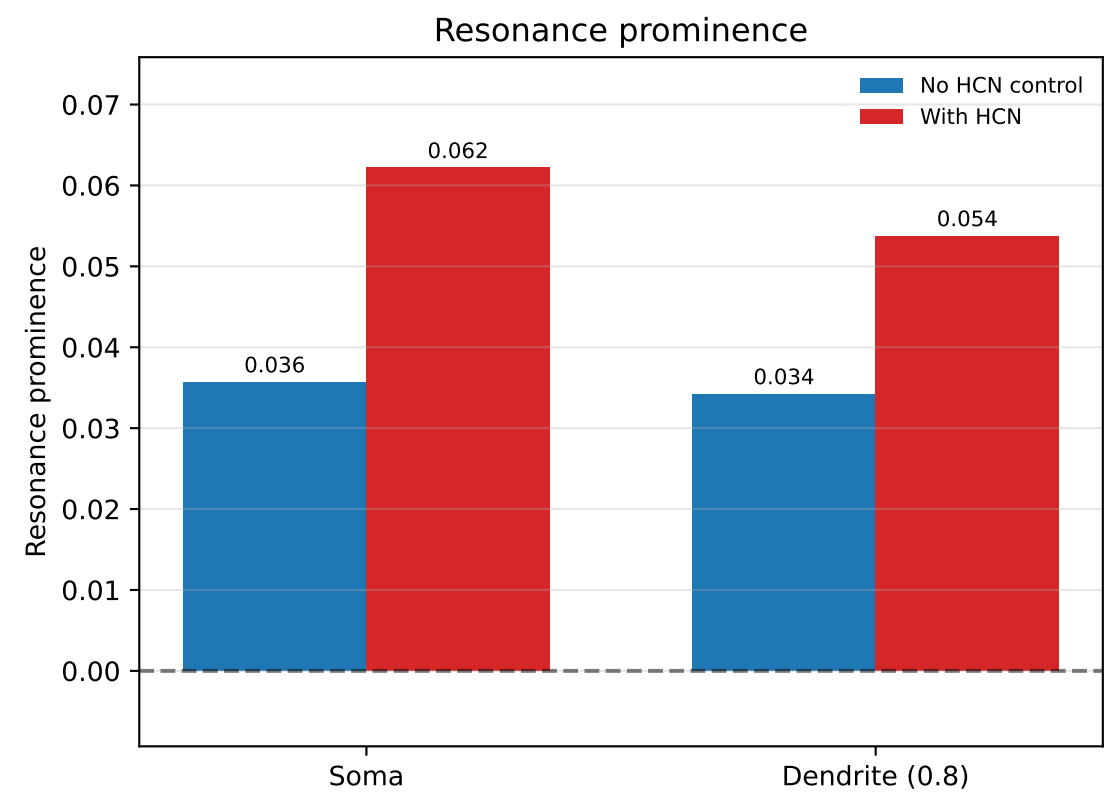
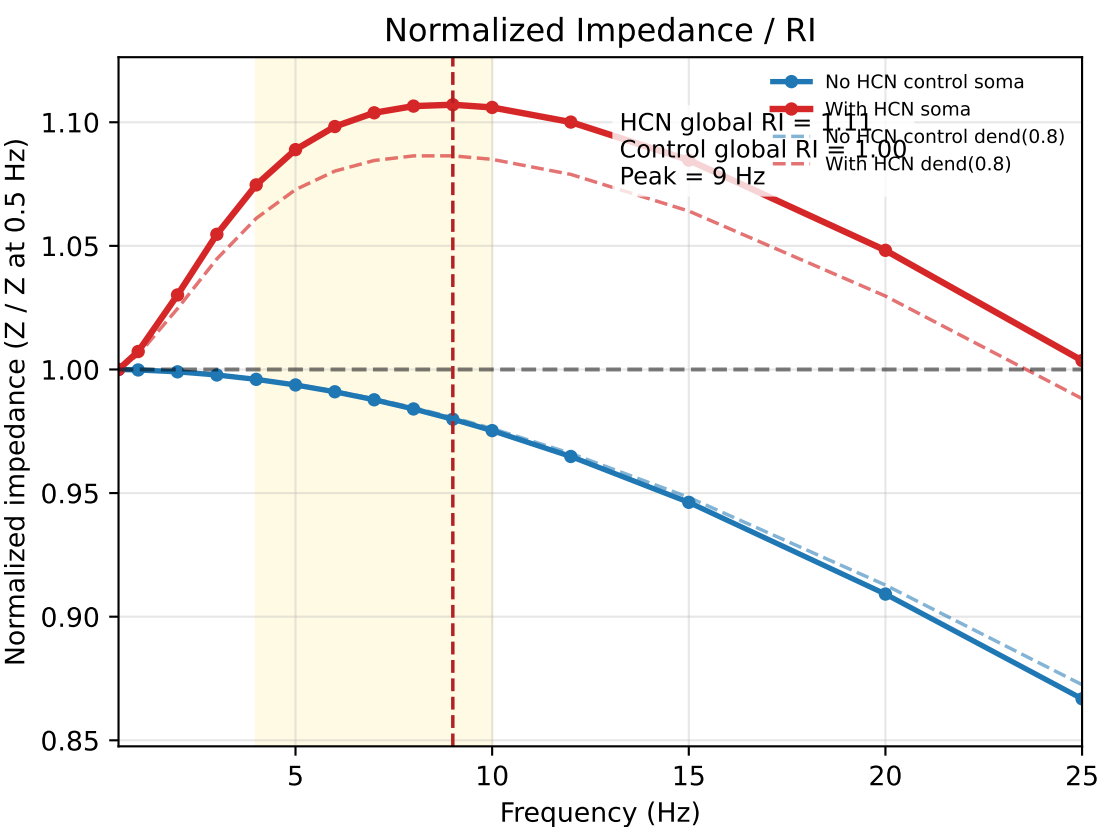
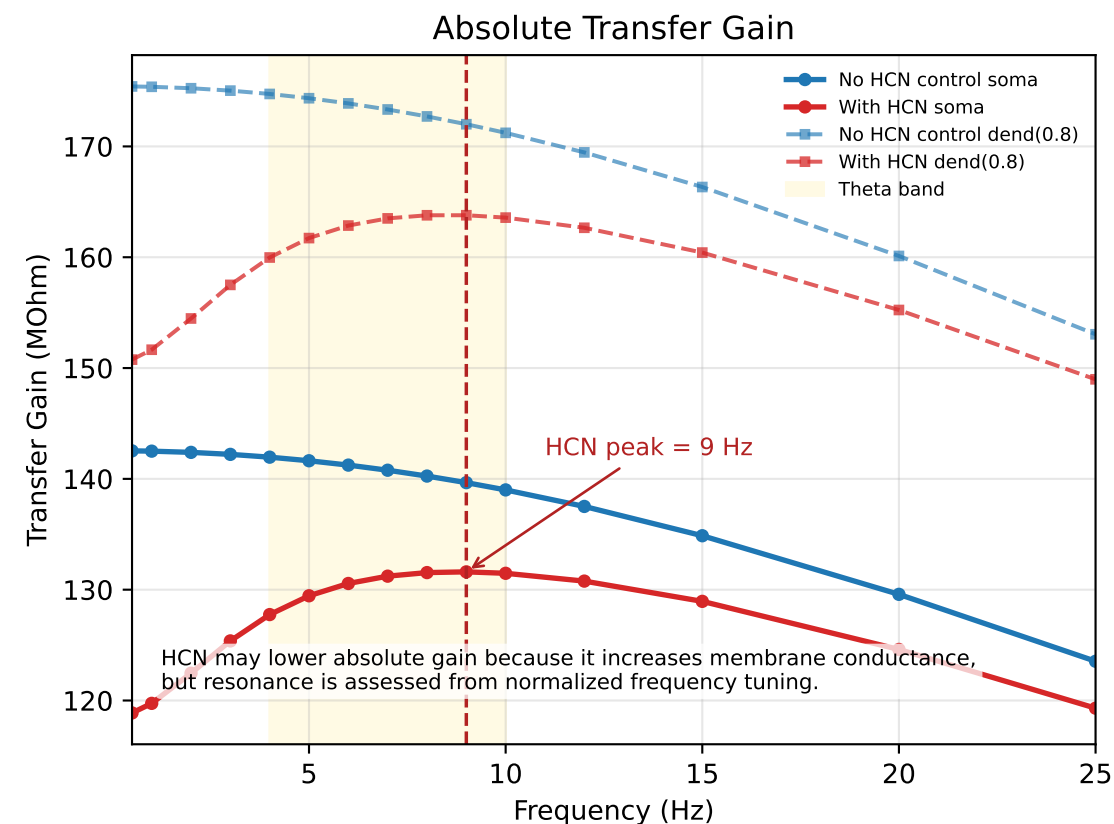
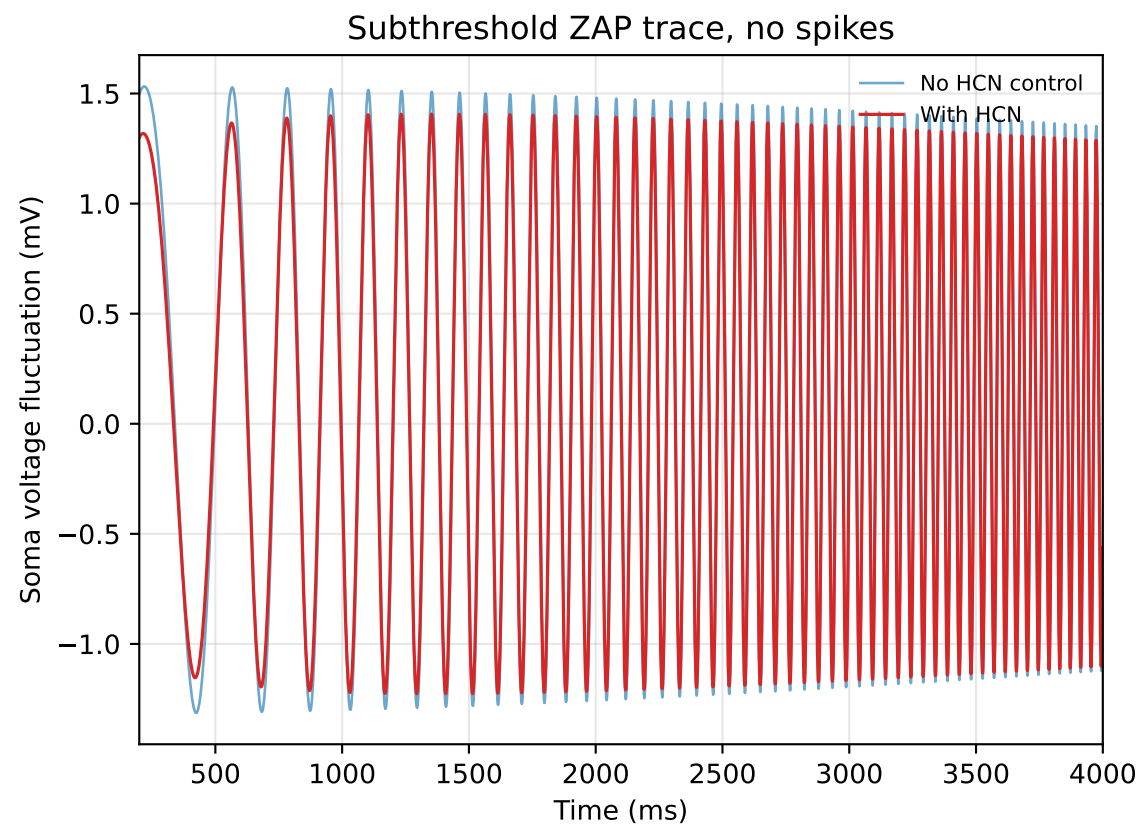
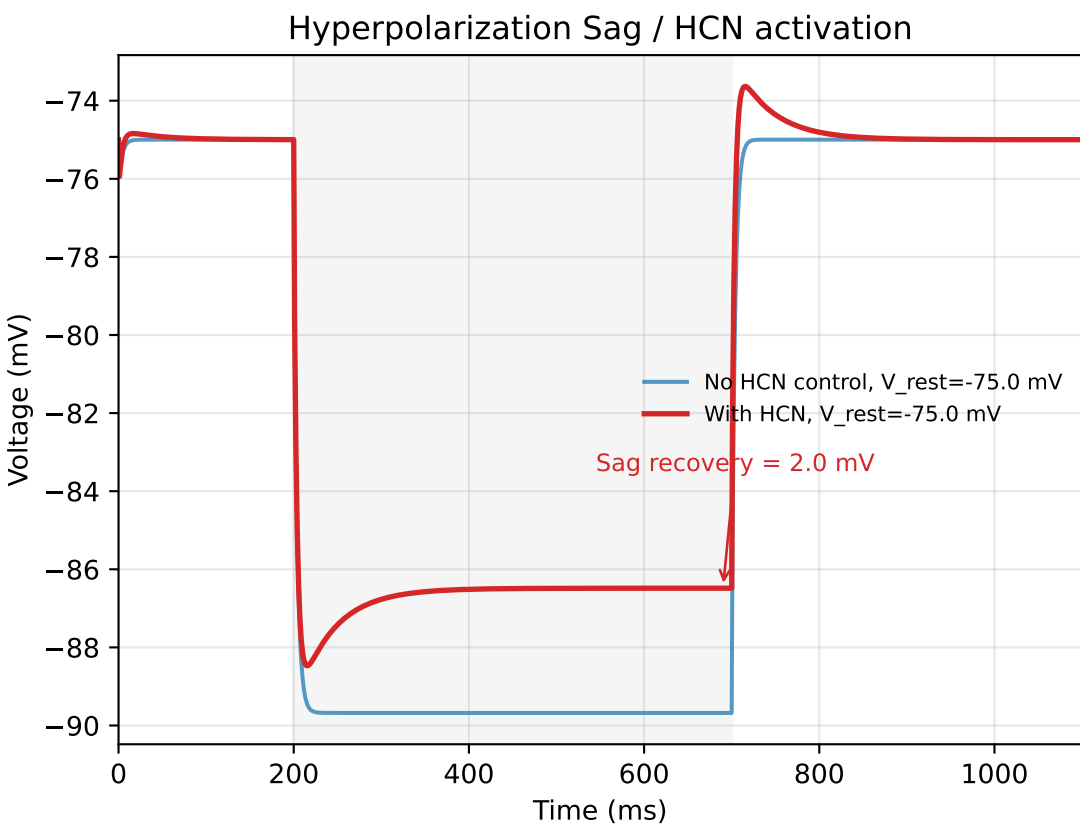


HCN/Ih reshapes subthreshold membrane frequency tuning toward resonance



Model:

- Soma: pas + hh1 Na/K +/- Ih
- Dendrite: pas +/- Ih
- Input: distal current at dend(0.8)
- Recording: soma and dend(0.8)

Operating point:

- Operating V_m : -75.0 mV
- Holding: current clamp, not voltage clamp
- Fixed g_h : 0.0008 S/cm²
- Soma Ih density: 0.25 x g_h
- Resonance band: 4.0-10.0 Hz

Validation:

- HH spike validation: True, current=0.08 nA
- ZAP/sweep for resonance: no spikes

Results:

- Sag recovery: control=-0.00 mV, HCN=1.99 mV
- Absolute gain can be lower with HCN because Ih increases membrane conductance
- Normalized impedance: HCN global RI > control global RI
- HCN global peak: 9.0 Hz
- Resonance prominence: HCN > control at soma and dendrite

Conclusion:

HCN/Ih produces relative resonance under hyperpolarized subthreshold conditions. This is a normalized frequency-tuning effect, not an absolute voltage-amplitude amplification.